

MCS-26 - M1101 - LAB 01 - CAESAR CIPHER

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Exercise 1.1 - Encrypt and decrypt

a) Encrypt 'cryptography is fun' using $k = 7$.

Numeric calculations for each letter:

c (2) + 7 = 9 % 26 = 9 (j)
r (17) + 7 = 24 % 26 = 24 (y)
y (24) + 7 = 31 % 26 = 5 (f)
p (15) + 7 = 22 % 26 = 22 (w)
t (19) + 7 = 26 % 26 = 0 (a)
o (14) + 7 = 21 % 26 = 21 (v)
g (6) + 7 = 13 % 26 = 13 (n)
r (17) + 7 = 24 % 26 = 24 (y)
a (0) + 7 = 7 % 26 = 7 (h)
p (15) + 7 = 22 % 26 = 22 (w)
h (7) + 7 = 14 % 26 = 14 (o)
y (24) + 7 = 31 % 26 = 5 (f)
i (8) + 7 = 15 % 26 = 15 (p)
s (18) + 7 = 25 % 26 = 25 (z)
f (5) + 7 = 12 % 26 = 12 (m)
u (20) + 7 = 27 % 26 = 1 (b)
n (13) + 7 = 20 % 26 = 20 (u)

Resulting ciphertext (spaces preserved):

jyfwavnyhwof pz mbu

b) Decrypt the result and verify you get the original text back.

By running `decrypt("jyfwavnyhwof pz mbu", 7)`, the cipher shifts each letter back by 7, which correctly yields the plaintext:

cryptography is fun

c) Repeat with $k = 19$ (rot-19) and verify symmetry.

By encrypting the ciphertext "jyfwavnyhwof pz mbu" with $k = 19$, we get "cryptography is fun". This verifies symmetry because $7 + 19 = 26$. A forward shift of 19 is mathematically equivalent to a backward shift of 7 (mod 26).

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Exercise 1.2 - Brute-force attack

Ciphertext to break:

MYXQVBKX ZKBDEOKX LOXGXXKVB0BK

a) Run the brute-force function on it and identify the key.

By running the brute_force function and minimizing the chi-squared score against English letter frequencies, we identify the key.

Recovered Key: 10

b) What is the plaintext?

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c) How many operations did the brute-force require? Would this be feasible by hand?

The brute-force attack requires 25 operations (trying all non-trivial shift values from 1 to 25) and observing which plaintext makes sense or has the lowest chi-squared score. Since 25 shifts is a very small number, this is entirely feasible to do by hand (especially if only decrypting the first few characters to check for English words).

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Exercise 1.3 - Frequency analysis

a) Encrypted text with a key of choice.

We used a 1299-character excerpt from the Wikipedia page on Cryptography and encrypted it with a key of 12.

b) Plot the letter-frequency histogram of the ciphertext.

The plot was successfully generated by the script and saved as 'caesar_freq.pdf'.

c) Use `find_key_frequency()` to recover the key automatically.

The algorithm successfully recovered the key: 12

d) Report the recovered key and the chi-squared scores for all 26 candidates.

Recovered Key: 12

Chi-squared scores:

0	2456.7605
1	1160.3936
2	833.4547
3	920.8551
4	1848.9461
5	769.9516
6	1729.7959
7	1378.4573
8	680.4035
9	745.7314
10	952.4928
11	798.5241
12	12.7510 <-- best
13	1415.5140
14	464.3837
15	1709.7229
16	760.4735
17	2944.8975
18	400.5273
19	1261.5629
20	884.8110
21	1270.1510
22	1045.9697
23	473.5900
24	563.1118
25	428.6308

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Exercise 1.4 - Security evaluation

Property	Caesar Cipher
1. Key space size	26 (or 25 non-trivial)
2. Time to brute-force (at 10^9 keys/sec)	2.6×10^{-8} seconds
3. Vulnerable to frequency analysis?	Yes
4. Index of Coincidence (ciphertext vs English)	Same (~ 0.065)
5. Attack type that breaks it	Frequency Analysis / Brute-Force